Question 1

R (A, B, C, D)

A → B
C → D

Decomposition:
R1 (A, B)
R2 (C, D)

Is this decomposition dependency preserving and lossless join?

Question 2

$R = \{A, B, C, D, E, F, G, H, I, J\}$

$F = \{$
$\{A, B\} \rightarrow \{C\}$
$\{A\} \rightarrow \{D, E\}$
$\{B\} \rightarrow \{F\}$
$\{F\} \rightarrow \{G, H\}$
$\{D\} \rightarrow \{I, J\}$
$\}$

Find the key of R.
Decompose it up to 3NF.

Question 3

Book (Book-title, Aname, Book-type, Price, Affil, Publisher)

Functional Dependencies:
Book-title → Publisher, Book-type
Book-type → Price
Aname → Affil

Question: Which normal form is the relation? Explain.
Decompose it up to 3NF.

Question 4

Three transactions T1, T2, T3 and schedule S1, S2.
Draw serializability graph.

T1 : r1(X), r1(Z), w1(X)
T2 : r2(Z), r2(Y)
T3 : r3(X), r3(Y), w3(Y)

S1 : r1(X); r2(Z); r1(Z); r3(X); r3(Y); w1(X); w3(Y); r2 (Y); w2(Z); w2(Y)

S2 : r1(X); r2(Z); r3(X); r1(Z); r2(Y); r3(Y); w1(X); w2(Z); w3(Y); w2(Y)